

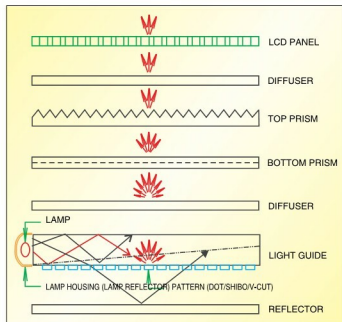


Fig. 3: Backlight arrangement

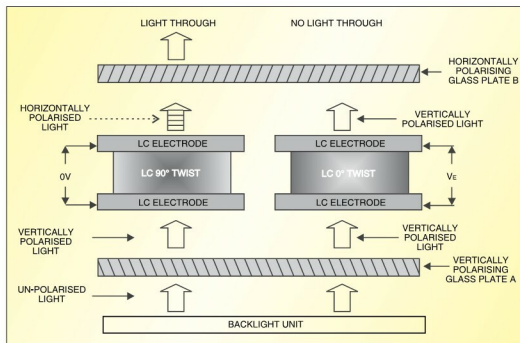


Fig. 4: Operation of LCD

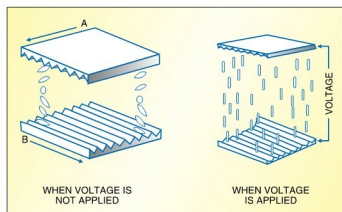


Fig. 5: Liquid crystal molecule alignment

between ITO electrodes and light is allowed to pass through with molecules twisted by 90°, changing the polarisation of light from vertical to horizontal. When an electric field is applied between ITO electrodes, liquid-crystal molecules get aligned by removing the 90° twist and light is blocked. When a smaller voltage is applied, the twist is less than 90° and lower-intensity light is passed. Thus the ap-

LCD uses one transistor for each row and each column. Each transistor is switched on in sequence and the column transistors are turned on/off according to the image to be displayed. When a transistor is turned off, the respective element or pixel loses its charge and blocks the passage of light. Because of this scanning and the slow response (response time greater than 150 ms), passive displays are not very bright and have low contrast ratio. The addressing technique causes crosstalk, which results in blurred images as non-selected pixels are driven through secondary voltage.

An active-matrix LCD uses a separate transistor for each unit pixel. This allows the pixels to stay 'on' longer and thus produce a brighter image. Therefore active-matrix LCDs are used in computer monitors, laptops, TVs, mobile screens, etc.

## Driving methods for TFT-LCDs

There are two ways to produce images on a liquid-crystal display: segment driving and matrix driving.

The segment-driving method is mostly used for simple displays, such as segment-type LCDs that are directly connected to the drivers.

The matrix-driving method is used for high-resolution displays. In this method, displays can be either statically driven or dynamically driven. In the static, or direct, driving method, each pixel is individually wired to the driver. But, as the number of pixels increases, wiring becomes complex. In the dynamic driving method, a voltage is applied at the intersections of specific horizontal-scanning electrodes and specific vertical signal electrodes in time-division multiplexing mode using pulse drive.

## Equivalent circuit of a unit pixel

A liquid-crystal capacitor ( $C_{LC}$ ) and a storage capacitor ( $C_s$ ) are connected

plied electric field determines the luminance of light passing through the liquid crystal and various greyscales can be obtained.

## Types of TFT-LCDs

There are two types of TFT-LCDs: passive-matrix and active-matrix.

A passive-matrix